

Homework 2

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Problem 1

Suppose that x, y are distinct points in a metric space (X, d) and let $\epsilon = \frac{d(x, y)}{2}$.
Prove that $B_\epsilon(x)$ and $B_\epsilon(y)$ are disjoint

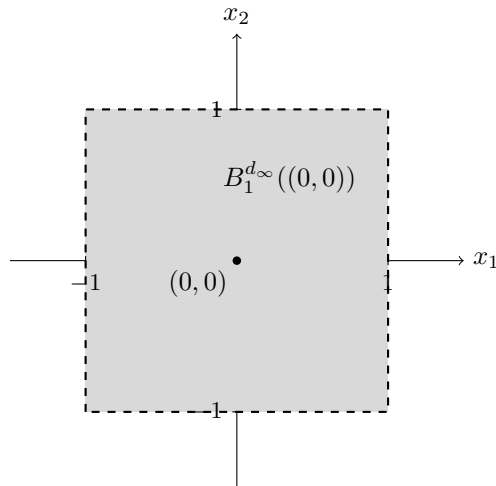
Solution. Assume there exists $v \in B_\epsilon(x) \cap B_\epsilon(y)$
Then $d(x, v) < \frac{d(x, y)}{2}$ and $d(y, v) < \frac{d(x, y)}{2}$

$$\begin{aligned}d(x, v) + d(y, v) &< d(x, y) \\d(x, y) &\leq d(x, v) + d(v, y) < d(x, y) \\d(x, y) &< d(x, y)\end{aligned}$$

A contradiction, hence the intersection of the two balls is empty, meaning they are disjoint □

Problem 2

Sketch the open ball $B_1^{d_\infty}((0, 0))$ in $\mathbb{R}^2$



Solution.

□

Problem 3

Prove that a subset of a metric space is open iff it is a union of open balls

Solution. Let (X, d) be a metric space and $U \subset X$
($\Rightarrow$)

Assume U is open

Then $\forall x \in U \exists \epsilon_x > 0$ s.t. $B_{\epsilon_x}(x) \subset U$

Then looking at $\cup_{x \in U} B_{\epsilon_x}(x)$

If $u \in U$, then certainly $u \in B_{\epsilon_u}(u)$, so $u \in \cup_{x \in U} B_{\epsilon_x}(x)$

If $u \in \cup_{x \in U} B_{\epsilon_x}(x)$, then $u \in B_{\epsilon_x}(x)$ for some x , which is a subset of U , so $u \in U$

Thus $U = \cup_{x \in U} B_{\epsilon_x}(x)$

($\Leftarrow$)

Lemma 1. Let (X, d) be a metric space
Any open ball is an open set

Proof. Let $\epsilon > 0$ and $x_0 \in X$ and consider $B_\epsilon(x_0) \subset X$

Let $x \in B_\epsilon(x_0)$, choose $\epsilon_x = \epsilon - d(x, x_0)$

Consider $B_{\epsilon_x}(x)$

If $y \in B_{\epsilon_x}(x)$ then $d(x, y) < \epsilon_x = \epsilon - d(x, x_0)$

$\Rightarrow d(y, x) + d(x, x_0) < \epsilon \Rightarrow d(y, x_0) < \epsilon$ by the triangle inequality.

Thus $y \in B_\epsilon(x_0)$ and then $B_{\epsilon_x}(x) \subset B_\epsilon(x_0)$

Thus $B_\epsilon(x_0)$ is an open set □

Let $\cup_{a \in A} B_{\epsilon_a}(a)$ be a union of open balls in X

From the above lemma $\cup_{a \in A} B_{\epsilon_a}(a)$ is also a union of open sets.

Then from lecture the union of opens sets is also open, so $\cup_{a \in A} B_{\epsilon_a}(a)$ is an open set □

Problem 4

Suppose that in a metric space X we have $B_r(x) = B_s(y)$ for some $x, y \in X$ and some positive real numbers r, s

Is $x = y$? and is $r = s$?

Solution. It is not true in general that $x = y$ and $r = s$

Consider the metric space of $(\mathbb{R}^n, d)$ for any n , where $d(x, y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$

Now for any $r > 1$ we have that $B_r(x) = \mathbb{R}^n$ for all $x \in \mathbb{R}^n$

For $n = 1$, we can take $x = 1$ and $y = 2$, $r = 2$ and $s = 3$, we have $B_r(x) = B_s(y) = \mathbb{R}$ while $x \neq y$ and $r \neq s$ □

Problem 5

Which of the following sets are closed in $\mathbb{R}$?

- $[1, \infty)$
- $\mathbb{R} \setminus \mathbb{Q}$
- $\left\{ \frac{n}{n+1} \mid n \in \mathbb{N} \right\}$
- $\left\{ \frac{1}{n} \mid n \in \mathbb{N}, n \geq 2 \right\} \cup \{0, 1, 2\}$

Solution. 1.

To check whether $[1, \infty)$ is closed, we need to see whether $\mathbb{R} \setminus [1, \infty) = (-\infty, 1)$ is open.

For $(-\infty, 1)$ to be open, we need $B_\epsilon(x) \subset (-\infty, 1)$ for all $x \in (-\infty, 1)$

Let $x \in (-\infty, 1)$, choose $\epsilon = \frac{|x-1|}{2}$

If $y \in B_\epsilon(x)$, then $|x - y| < \epsilon = \frac{|x-1|}{2}$

So,

$$-\frac{|x-1|}{2} < x-y$$
$$y < \frac{|x-1|}{2} + x$$

Since $x \in (-\infty, 1) \Rightarrow x - 1 \leq 0 \Rightarrow |x - 1| = 1 - x$

Then $y < x - \frac{x-1}{2} = \frac{1}{2}(x+1)$

Let $f: \mathbb{R} \rightarrow \mathbb{R}: f(x) = \frac{1}{2}(x+1)$

Clearly f is increasing since $f' > 0$

Thus $\forall x \in (-\infty, 1) f(x) < f(1) = 1$
 So $y < 1$ so $B_\epsilon(x) \subset (-\infty, 1)$
 Hence $(-\infty, 1)$ is open, so $[1, \infty)$ is closed □

Solution. 2.

Lemma 2. *Between any two rational numbers a, b , $a < b$ there exists an irrational number*

Proof. Let $r = a + \frac{b-a}{\sqrt{2}}$
 Clearly $\frac{b-a}{\sqrt{2}} > 0$ so $a < r$, and also $b - a > r - a = \frac{b-a}{\sqrt{2}}$
 Combining these we get $a < r < b$

Assume r is rational and can therefore be written as the ratio of two coprime integers $p, q, q \neq 0$

$$r = a + \frac{b-a}{\sqrt{2}}$$

$$\Rightarrow \sqrt{2} = \frac{b-a}{r-a}$$

Since $r \neq a$ we have $\sqrt{2}$ as the ratio of rationals, since the sums of rationals are rationals, which is rational
 A contradiction to the fact that $\sqrt{2}$ is known to be irrational
 Hence r is irrational □

Assume for the sake of contradiction that $\mathbb{R} \setminus \mathbb{Q}$ is closed then $\mathbb{R} \setminus (\mathbb{R} \setminus \mathbb{Q}) = \mathbb{Q}$ is open
 But we know that $\mathbb{Q}$ is not open
 Let $q \in \mathbb{Q}$ and $\epsilon > 0$

$$B_\epsilon(q) = \{x \in \mathbb{R} \mid |x - q| < \epsilon\}$$

We can find two rationals in $B_\epsilon(q)$, for example $q \pm \frac{\epsilon}{2}$, then from the above lemma there exists some $r \in \mathbb{R} \setminus \mathbb{Q}$
 such that $r \in B_\epsilon(q)$
 Hence $B_\epsilon(q) \not\subset \mathbb{Q}$
 Hence $\mathbb{Q}$ is not open
 Hence $\mathbb{R} \setminus \mathbb{Q}$ is not closed □

Solution. 3.

If $A = \left\{1 - \frac{1}{n+1} \mid n \in \mathbb{N}\right\}$ is closed then $\bar{A} = A$

Clearly $1 \notin A$ since $\frac{1}{n+1} \neq 0$ for all $n \in \mathbb{N}$

But we can create a sequence x_n with $x_n \in A$ for all $n \in \mathbb{N}$, which converges to 1, $\left\{x_n = 1 - \frac{1}{n+1} \mid n \in \mathbb{N}\right\}$.
 This obviously converges to 1, since $\frac{1}{n+1} \rightarrow 0$. Thus from the sequential characterization of the closure $1 \in \bar{A}$
 Since $1 \notin A$ and $1 \in \bar{A}$ this implies $\bar{A} \neq A$, and hence A is not closed □

Solution. 4.

First we will find the limit points of $A = \left\{\frac{1}{n} \mid n \in \mathbb{N}\right\} \cup \{0, 1, 2\}$

It is clear from the definition of a limit point that the right hand side of the union has no limit points, since for $\epsilon = \frac{1}{2}$ $B_\epsilon(x) \setminus \{x\} \cap \{0, 1, 2\} = \emptyset$ for all $x \in \{0, 1, 2\}$

We can also see that 0 is a limit point of the left hand side, consider the sequence $\left\{x_n = \frac{1}{n}\right\}_{n=2}^\infty$, this is clearly convergent to 0 and with each of the $x_n \in A$ and $x_n \neq 0$ for all $n \in \mathbb{N}$

Consider any $a < 0$, every element of A is greater or equal to zero, so $B_{\frac{|a|}{2}}(a) = \left(\frac{3a}{2}, \frac{a}{2}\right)$ does not intersect with A , so no $a < 0$ is a limit point

Now consider any $a > 0$. $B_{\frac{a}{2}}(a) = \left(\frac{a}{2}, \frac{3a}{2}\right)$, a point $\frac{1}{n} \in A, n \in \mathbb{N}$ lies in the ball iff $\frac{1}{n} > \frac{a}{2} \Rightarrow n < \frac{2}{a}$.
 Only a finite number of n can satisfy this, hence the ball contains a finite amount of elements in A . Let ϵ be the minimum of all of the distances between the points of a in $B_{\frac{a}{2}}(a) \setminus \{a\}$ to a . By construction, $B_\epsilon(a) \setminus \{a\} \cap A = \emptyset$, so no $a > 0$ is a limit point.

Hence 0 is the only limit point of A

But $0 \in A$, so $\bar{A} = A \cup A' = A$, hence A is closed □